

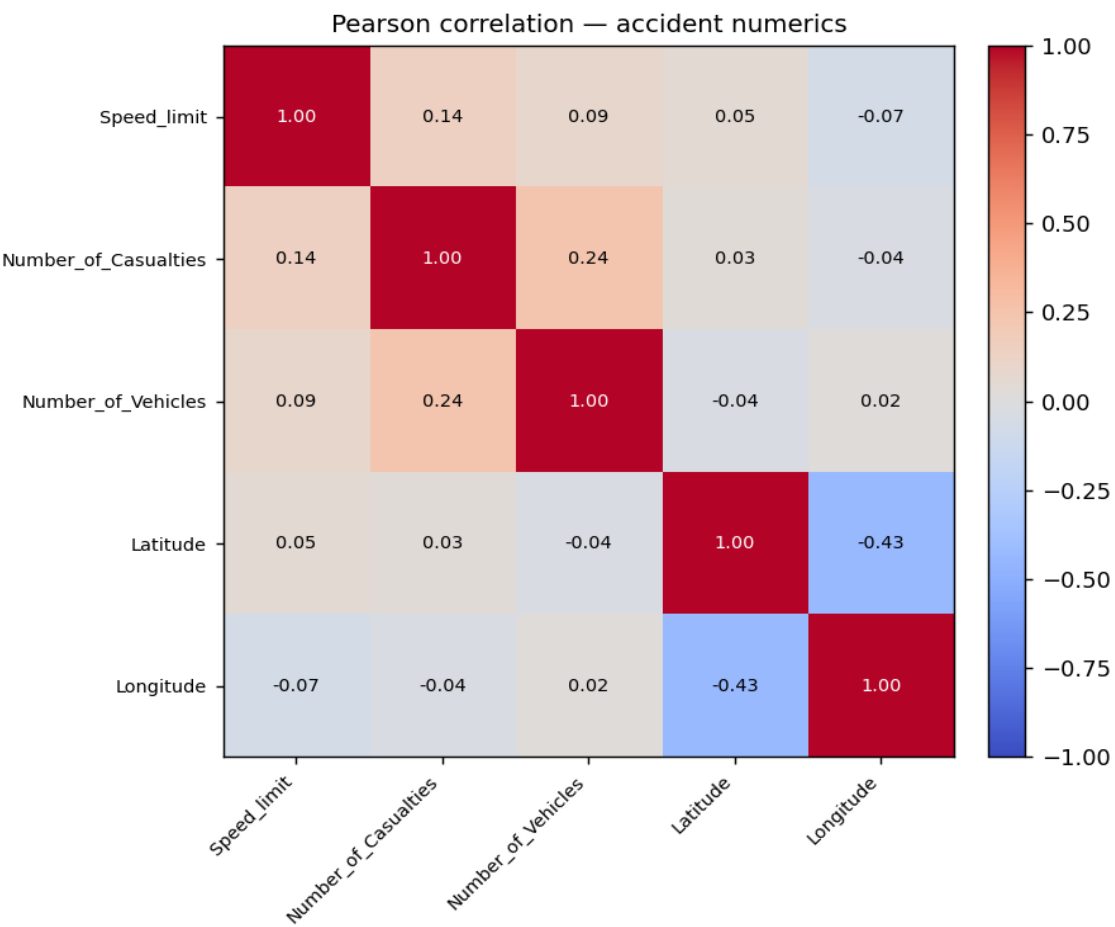
UK Road Accidents — Data Validation Report

Generated: 2026-05-02T05:35:17.191499

Total Issues Found: 18

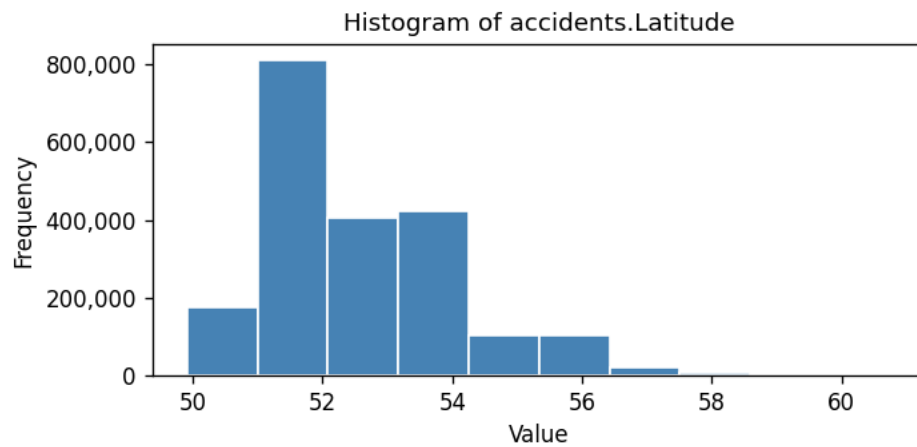
Dimension	Issues
ACCURACY	2
COMPLETENESS	3
JOIN_INTEGRITY	2
WEATHER_DISTRIBUTION	11

Correlation Heatmap

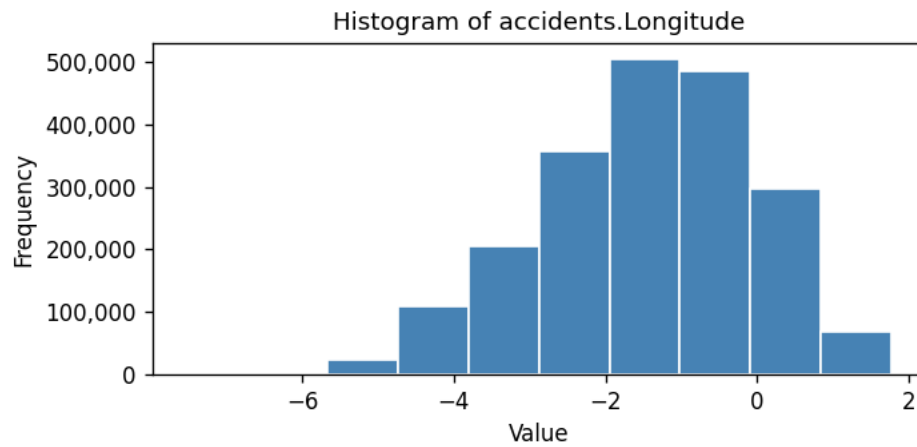


Distribution Histograms

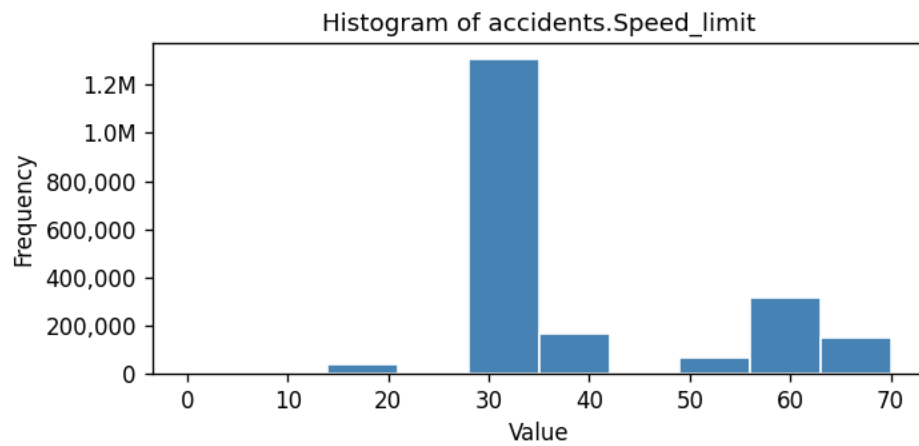
accidents.Latitude



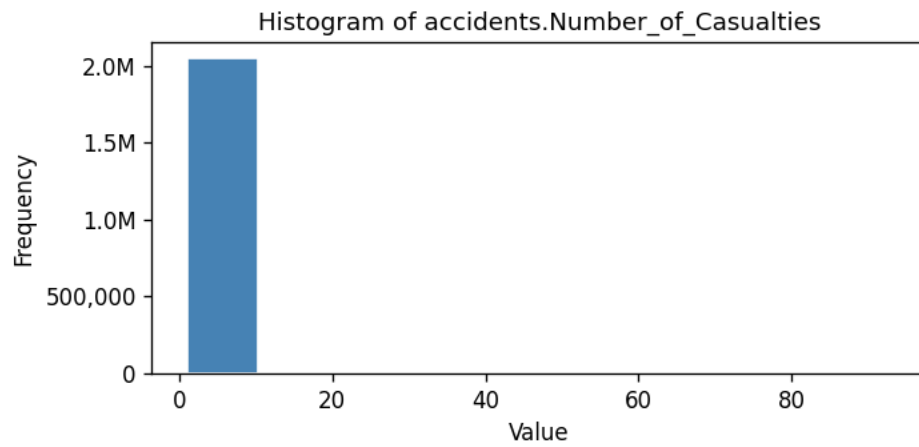
accidents.Longitude



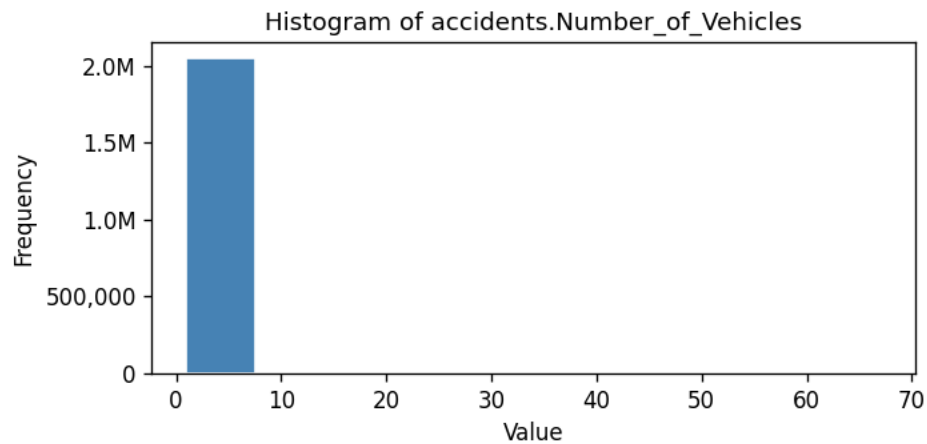
accidents.Speed_limit



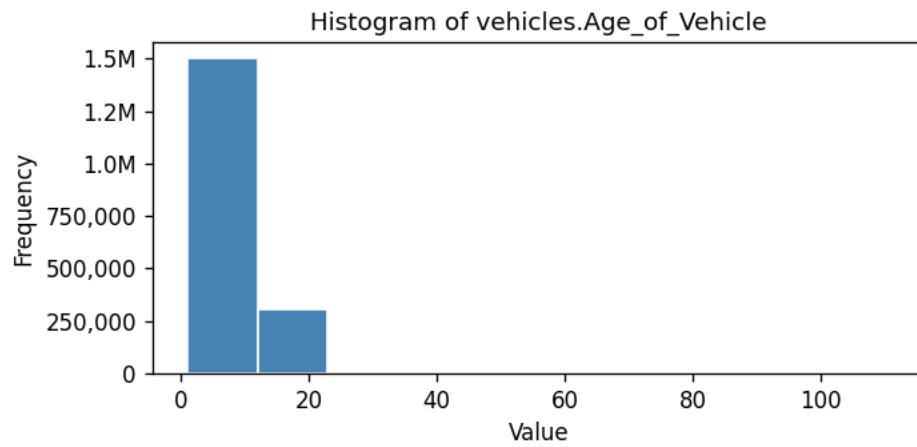
accidents.Number_of_Casualties



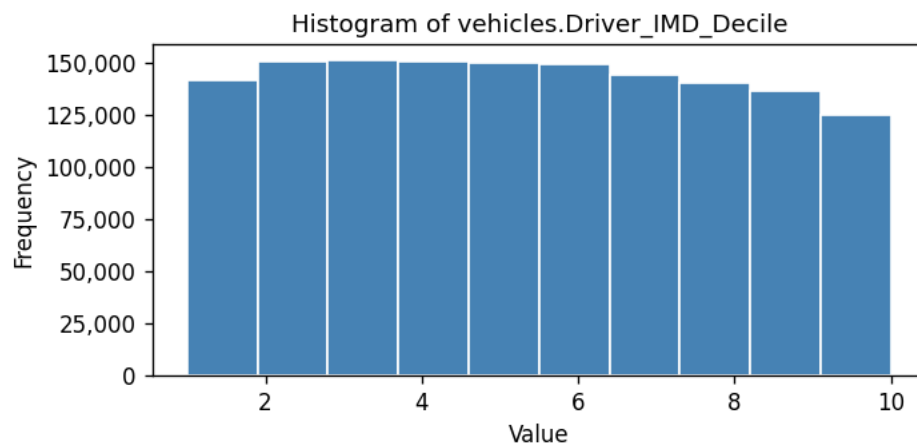
accidents.Number_of_Vehicles



vehicles.Age_of_Vehicle



vehicles.Driver_IMD_Decile



Full Validation Log

```
=====
UK Road Accidents – Data Validation Report
2026-05-02 05:35:20
=====
Loaded accidents : 2,047,256 rows x 34 columns
Loaded vehicles : 2,177,205 rows x 24 columns
-----
DIMENSION 1 – ACCURACY
-----
[PASS] Accident_Severity – all values valid (Slight/Serious/Fatal)
[ISSUE] 36 rows have Speed_limit not in UK legal set {50, 20, 70, 40, 60, 30}
[PASS] Latitude – all within UK bounds (49-61)
[PASS] Longitude – all within UK bounds (-8 to 2)
[PASS] Number_of_Vehicles – all values >= 1
[PASS] Age_of_Vehicle – no negative values
[INFO] Age_Band_of_Driver has 12 unique categories
Sample values: ['26 - 35', '66 - 75', '36 - 45', '46 - 55', '21 - 25', 'Data missing or out of range', '16 - 20', '
56 - 65', 'Over 75', '11 - 15']
[PASS] Day_of_Week – all values valid
[ISSUE] 76,119 vehicle records have invalid Sex_of_Driver values
-----
DIMENSION 2 – CONSISTENCY
-----
[PASS] Date column – parses correctly as datetime
[PASS] Year column – matches year extracted from Date in all rows
[WARN] 2 Date/Day_of_Week mismatches found
[PASS] Time column – all values follow HH:MM format
[PASS] Year – consistent across both tables for all matched accidents
-----
DIMENSION 3 – COMPLETENESS
-----
[ACCIDENTS]
Column Missing %
-----
Carriageway_Hazards 2,007,807 98.1%
Special_Conditions_at_Site 1,995,137 97.5%
2nd_Road_Class 844,272 41.2%
LSOA_of_Accident_Location 144,953 7.1%
2nd_Road_Number 17,593 0.9%
Pedestrian_Crossing-Physical_Facilities 3,560 0.2%
Pedestrian_Crossing-Human_Control 2,920 0.1%
Did_Police_Officer_Attend_Scene_of_Accident 278 0.0%
Latitude 174 0.0% [!] REQUIRED
Location_Easting_OSGR 164 0.0%
Location_Northing_OSGR 164 0.0%
Longitude 175 0.0% [!] REQUIRED
[ISSUE] [ACCIDENTS] CRITICAL: 'Latitude' has 174 nulls (0.01%). These rows must be dropped.
[ISSUE] [ACCIDENTS] CRITICAL: 'Longitude' has 175 nulls (0.01%). These rows must be dropped.
[ISSUE] [ACCIDENTS] CRITICAL: 'Time' has 156 nulls (0.01%). These rows must be dropped.
[ADVICE] 'Carriageway_Hazards' is 98.07% missing. Consider dropping this feature.
[ADVICE] 'Special_Conditions_at_Site' is 97.45% missing. Consider dropping this feature.
[VEHICLES]
Column Missing %
-----
Hit_Object_in_Carriageway 2,087,824 95.9%
Hit_Object_off_Carriageway 1,989,741 91.4%
Skidding_and_Overturning 1,898,208 87.2%
Driver_IMD_Decile 734,812 33.8%
Age_of_Vehicle 358,149 16.4%
model 325,331 14.9%
Engine_Capacity_.CC. 265,861 12.2%
Propulsion_Code 245,843 11.3%
make 110,845 5.1%
Vehicle_Location.Restricted_Lane 1,317 0.1%
[PASS] All strictly required columns in VEHICLES are 100% complete.
[ADVICE] 'Hit_Object_in_Carriageway' is 95.89% missing. Consider dropping this feature.
```



```
[ 19.4 - 28.6] 56 ( 0.0%) █
[ 28.6 - 37.8] 17 ( 0.0%) █
[ 37.8 - 47.0] 15 ( 0.0%) █
[ 47.0 - 56.2] 5 ( 0.0%) █
[ 56.2 - 65.4] 3 ( 0.0%) █
[ 65.4 - 74.6] 2 ( 0.0%) █
[ 74.6 - 83.8] 0 ( 0.0%) █
[ 83.8 - 93.0] 3 ( 0.0%) █
accidents.Number_of_Vehicles 1.0 67.0 1.83 2.00 0.71 2.16 53.08 29
Histogram (accidents.Number_of_Vehicles):
[ 1.0 - 7.6] 2,046,478 (100.0%) ██████████
[ 7.6 - 14.2] 749 ( 0.0%) █
[ 14.2 - 20.8] 20 ( 0.0%) █
[ 20.8 - 27.4] 3 ( 0.0%) █
[ 27.4 - 34.0] 3 ( 0.0%) █
[ 34.0 - 40.6] 2 ( 0.0%) █
[ 40.6 - 47.2] 0 ( 0.0%) █
[ 47.2 - 53.8] 0 ( 0.0%) █
[ 53.8 - 60.4] 0 ( 0.0%) █
[ 60.4 - 67.0] 1 ( 0.0%) █
vehicles.Age_of_Vehicle 1.0 111.0 7.11 7.00 4.73 1.22 5.72 88
Histogram (vehicles.Age_of_Vehicle):
[ 1.0 - 12.0] 1,500,285 ( 82.5%) ██████████
[ 12.0 - 23.0] 308,862 ( 17.0%) ████████
[ 23.0 - 34.0] 8,207 ( 0.5%) █
[ 34.0 - 45.0] 1,075 ( 0.1%) █
[ 45.0 - 56.0] 460 ( 0.0%) █
[ 56.0 - 67.0] 109 ( 0.0%) █
[ 67.0 - 78.0] 36 ( 0.0%) █
[ 78.0 - 89.0] 15 ( 0.0%) █
[ 89.0 - 100.0] 4 ( 0.0%) █
[ 100.0 - 111.0] 3 ( 0.0%) █
vehicles.Driver_IMD_Decile 1.0 10.0 5.39 5.00 2.82 0.05 -1.19 10
Histogram (vehicles.Driver_IMD_Decile):
[ 1.0 - 1.9] 141,939 ( 9.8%) ██████
[ 1.9 - 2.8] 151,065 ( 10.5%) ██████
[ 2.8 - 3.7] 151,377 ( 10.5%) ██████
[ 3.7 - 4.6] 151,068 ( 10.5%) ██████
[ 4.6 - 5.5] 150,367 ( 10.4%) ██████
[ 5.5 - 6.4] 149,351 ( 10.4%) ██████
[ 6.4 - 7.3] 144,579 ( 10.0%) ██████
[ 7.3 - 8.2] 140,578 ( 9.7%) ██████
[ 8.2 - 9.1] 136,802 ( 9.5%) ██████
[ 9.1 - 10.0] 125,267 ( 8.7%) ██████
[SKIP] Column 'Number_of_Casualties' not found
[SKIP] Column 'Number_of_Occupants' not found
KS Test vs Normal Distribution:
Column KS stat p-value Interpretation
-----
accidents.Latitude 0.1305 0.000000 NOT normally distributed (p < 0.05) – use Spe
accidents.Longitude 0.0882 0.000000 NOT normally distributed (p < 0.05) – use Spe
accidents.Speed_limit 0.3998 0.000000 NOT normally distributed (p < 0.05) – use Spe
accidents.Number_of_Casualties 0.4400 0.000000 NOT normally distributed (p < 0.05) – use Spe
accidents.Number_of_Vehicles 0.3028 0.000000 NOT normally distributed (p < 0.05) – use Spe
vehicles.Age_of_Vehicle 0.0996 0.000000 NOT normally distributed (p < 0.05) – use Spe
vehicles.Driver_IMD_Decile 0.1064 0.000000 NOT normally distributed (p < 0.05) – use Spe
-----
DIMENSION 7b – WEATHER COLUMN DISTRIBUTION PROFILE
-----
Column Missing% Min Max Mean Median Std Skew Kurt
-----
[ISSUE] [WEATHER] 'temp' is 48.8% missing (>20.0% threshold)
temp 48.8% -13.20 27.90 10.66 10.90 5.25 -0.17 -0.50
[PASS] temp – all values within physical bounds [-30, 50]
Histogram (temp):
[ -13.20 - -9.09] 152 ( 0.0%) █
[ -9.09 - -4.98] 904 ( 0.1%) █
[ -4.98 - -0.87] 13,999 ( 1.0%) █
[ -0.87 - 3.24] 113,350 ( 8.2%) ██████
[ 3.24 - 7.35] 260,672 ( 18.7%) ██████████
```

[illegible]

```
KS=0.0735 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'pres' is 73.9% missing (>20.0% threshold)
pres 73.9% 955.20 1044.20 1014.84 1015.70 10.78 -0.46 0.43
[PASS] pres – all values within physical bounds [870, 1085]
```

Histogram (pres):

[illegible]

```
KS=0.0416 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'rhum' is 49.4% missing (>20.0% threshold)
rhum 49.4% 4.00 100.00 79.99 81.00 9.81 -0.44 -0.10
[PASS] rhum – all values within physical bounds [0, 100]
```

Histogram (rhum):

[illegible]

```
KS=0.0652 p=0.000000 → NOT normal (p<0.05) – use Spearman
[ISSUE] [WEATHER] 'wpgt' is 100.0% missing (>20.0% threshold)
wpgt 100.0% [SKIP – too few valid values]
[ISSUE] [WEATHER] 'tsun' is 100.0% missing (>20.0% threshold)
tsun 100.0% [SKIP – too few valid values]
[ISSUE] [WEATHER] 'cldc' is 85.6% missing (>20.0% threshold)
cldc 85.6% 0.00 8.00 5.09 5.00 1.80 -0.71 0.23
[PASS] cldc – all values within physical bounds [0, 9]
```

Histogram (cldc):

[0.00 - 0.80]	9,133 (2.3%)	■
[0.80 - 1.60]	8,549 (2.2%)	■
[1.60 - 2.40]	18,176 (4.6%)	■■
[2.40 - 3.20]	29,460 (7.5%)	■■■
[3.20 - 4.00]	0 (0.0%)	■
[4.00 - 4.80]	68,104 (17.4%)	■■■■■■■■■
[4.80 - 5.60]	71,745 (18.3%)	■■■■■■■■■■
[5.60 - 6.40]	96,445 (24.6%)	■■■■■■■■■■■■
[6.40 - 7.20]	69,912 (17.9%)	■■■■■■■■■
[7.20 - 8.00]	19,925 (5.1%)	■■■

KS=0.1647 p=0.000000 → NOT normal (p<0.05) – use Spearman

DIMENSION 8 – RELATIONSHIPS

Pearson Correlation Matrix:

```
Speed_limit Number_of_C Number_of_V Latitude Longitude
Speed_limit 1.000 0.143 0.088 0.052 -0.066
Number_of_Casualties 0.143 1.000 0.237 0.030 -0.038
Number_of_Vehicles 0.088 0.237 1.000 -0.039 0.019
Latitude 0.052 0.030 -0.039 1.000 -0.434
Longitude -0.066 -0.038 0.019 -0.434 1.000
```

Pearson vs Spearman (|diff| > 0.10 = non-linear relationship detected):
Pair Pearson Spearman |Diff| Note

Speed_limit x Number_of_Casualties 0.143 0.162 0.019
Speed_limit x Number_of_Vehicles 0.088 0.069 0.019
Speed_limit x Latitude 0.052 0.034 0.018
Speed_limit x Longitude -0.066 -0.062 0.004
Number_of_Casualties x Number_of_Vehicles 0.237 0.230 0.007
Number_of_Casualties x Latitude 0.030 0.044 0.014
Number_of_Casualties x Longitude -0.038 -0.046 0.008
Number_of_Vehicles x Latitude -0.039 -0.029 0.010
Number_of_Vehicles x Longitude 0.019 0.018 0.001
Latitude x Longitude -0.434 -0.411 0.023
[PASS] No high-correlation pairs found (threshold |r| > 0.85)
[SAVED] reports\correlation_heatmap.png

JOIN INTEGRITY (Accident_Index key across both tables)

Unique Accident_Index in accidents table : 2,047,256
Unique Accident_Index in vehicles table : 1,488,981
Keys matched (will survive inner join) : 1,389,724
[ISSUE] 657,532 accidents (32.1%) have no vehicle record – dropped on inner join, document in cleaning step
[ISSUE] 99,257 vehicle records (6.7%) have no matching accident – orphaned records, dropped on inner join
=====

VALIDATION COMPLETE

Total issues found across all 8 dimensions: 18

Issues by dimension:

[ACCURACY] 36 rows have Speed_limit not in UK legal set {50, 20, 70, 40, 60, 30}
[ACCURACY] 76,119 vehicle records have invalid Sex_of_Driver values
[COMPLETENESS] [ACCIDENTS] CRITICAL: 'Latitude' has 174 nulls (0.01%). These rows must be dropped.
[COMPLETENESS] [ACCIDENTS] CRITICAL: 'Longitude' has 175 nulls (0.01%). These rows must be dropped.
[COMPLETENESS] [ACCIDENTS] CRITICAL: 'Time' has 156 nulls (0.01%). These rows must be dropped.
[WEATHER_DISTRIBUTION] [WEATHER] 'temp' is 48.8% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'tmin' is 56.8% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'tmax' is 56.7% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'prcp' is 94.5% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'snwd' is 98.7% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'wspd' is 51.7% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'pres' is 73.9% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'rhum' is 49.4% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'wpgt' is 100.0% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'tsun' is 100.0% missing (>20.0% threshold)
[WEATHER_DISTRIBUTION] [WEATHER] 'cldc' is 85.6% missing (>20.0% threshold)
[JOIN_INTEGRITY] 657,532 accidents (32.1%) have no vehicle record – dropped on inner join, document in cleaning step
[JOIN_INTEGRITY] 99,257 vehicle records (6.7%) have no matching accident – orphaned records, dropped on inner join
[SAVED] reports\validation_report.json
[SAVED] reports\validation_summary.txt